



Quantum group consensus model based on linguistic Z-numbers and its application to meteorological disaster emergency

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ARTICLE INFO

Keywords:

Multi-attribute group decision making
Linguistic Z-numbers
Group consensus process
Deng entropy
Quantum probability theory

ABSTRACT

In the meteorological disaster emergency decision-making problem, the quantitative collection of decision information is complex and uncertain, and the mutual interference between decision makers affects group decision-making. To solve this problem, a quantum group consensus model based on linguistic Z-number is proposed. Firstly, the model uses linguistic Z-number to represent the evaluation information of decision makers, and provides feedback matrix to assist decision adjustment. In addition, the model also considers the effect of confidence on the efficiency of group consensus adjustment, and adjusts the weights of decision makers according to the distance between the final decision matrix and the group consensus matrix. Secondly, a maximum Deng entropy model is constructed to determine the attributes' weights in uncertain environments. On this basis, a quantum-like Bayesian network is constructed, the interference angle is measured by using Deng entropy, and the quantum probabilities of alternatives are calculated. Finally, the applicability and superiority of the model are verified through meteorological disaster case. Sensitivity and comparative analysis confirm the effectiveness of the model under different conditions. The results show that the model can improve group decision consensus under different confidence levels and multi-dimensional matrix input, and has strong robustness and wide applicability.

1. Introduction

In the face of highly uncertain and complex disasters, such as meteorological events, group decision-making (GDM) becomes crucial [1]. Meteorological disasters are extreme weather events caused by atmospheric conditions, leading to significant damage and disruption. These disasters include hurricanes, tornadoes, floods, droughts, and severe storms, often resulting in loss of life, property damage, and long-term socio-economic impacts. The frequency and intensity of such events have been increasing due to climate change, making effective emergency management and decision-making critical. Consequently, effective GDM leverages collective wisdom and resources to better respond to these risks and emergencies [2,3]. GDM involves multiple actors, including government agencies, community organizations, professional institutions, and the general public, who collaborate to share information and

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resources, developing and implementing strategies to minimize losses and protect lives and property. However, the knowledge limitations and subjective biases of decision-makers can affect decision quality. Additionally, accurate quantitative data is often difficult to obtain, making linguistic information essential for managing uncertainty [4,5]. To attempt to improve the usefulness of models in complicated situations, there has been a development in how linguistic information is represented, including the 2-tuple linguistic model [6,7], hesitant fuzzy linguistic term sets (HFLTS) [8] and probabilistic linguistic term sets (PLTS) [9]. However, these representations are not always effective. Linguistic Z-numbers (LZNs) can effectively combine qualitative description with quantitative analysis by providing fuzzy quantitative ranges, thus significantly improving the quality of decision making [10]. LZNs uses linguistic preference and fuzzy logic to successfully deal with uncertainty in the decision-making process [11], and is particularly suitable for dealing with the complexity and uncertainty of meteorological disaster information. As a result, LZNs is ideal for modeling data complexity in the field. Compared to other linguistic term sets, LZNs offers greater flexibility and accuracy, making it more adaptable in highly uncertain decision environments [12,13].

In group decision-making, achieving effective consensus is crucial [14]. To reach this goal, the decision-making process must incorporate a structured consensus process [15,16]. This process typically consists of two main stages: the measurement and quantification of consensus levels, and preference adjustment based on consensus rules [17]. Initially, decision makers evaluate and quantitatively analyze the group's initial consensus level, proposing specific preference adjustments to gradually align with the desired consensus level [18]. This mechanism enables the group decision-making process to address and resolve potential differences effectively while maintaining a high degree of coordination and consistency [19]. For example, a framework based on bilateral negotiation mechanisms can enhance group consensus through interactions among individual decision-makers, considering the limited compromise behavior of decision-makers during consensus building [20]. Additionally, the cost of adjustments in the consensus process directly impacts the efficiency and feasibility of reaching consensus. By minimizing adjustment costs, decision-makers can achieve consensus more effectively [21]. For instance, Fan et al. [22] proposed a consensus model for optimal resource allocation, accounting for resource allocation problems and utility functions, which supports mediators in achieving consensus under uncertain unit costs. Moreover, Sun et al. [23] proposed a theoretical framework to thwart the manipulation of weights in collective decision-making inside social networks. By employing the minimum adjustment feedback model and the maximum entropy principle, effective consensus was achieved and validated through numerical and simulation analyses. Therefore, by comprehensively considering the impact of consensus level measurement and quantification, preference adjustments based on consensus rules, and adjustment costs, decision makers can enhance the effectiveness and feasibility of group decision-making, thereby improving the quality and acceptance of decisions.

In the multi-attribute group decision-making (MAGDM) environment of uncertainty and complexity, determining the weight distribution of each attribute reasonably is an important factor to ensure the accuracy of decision results [24–26]. While the entropy weight method is widely used in this field, it has limitations in many scenarios. Although traditional Shannon entropy [27] is commonly employed for uncertainty measurement, it may be inadequate when dealing with complex uncertain information. By contrast, Deng entropy, a generalization of Shannon entropy, is a versatile tool for measuring uncertainty. It offers a promising approach to quantifying uncertainty and handling more ambiguous information. Designed to measure the disordered state of system information, Deng entropy extracts and utilizes information by maximizing entropy [28]. This capability is crucial in complex decision-making contexts, where accurately capturing and utilizing all available information is essential for effective decisions. Therefore, this paper employs Deng entropy to assign attributes' weights, aiming to better represent the credibility of evaluation information and expand the range of applications in uncertain contexts. This approach not only enhances the dependability of decision-making but also improves the retention and usage efficiency of information during the decision-making process, ultimately aiding in achieving superior decision results in dynamic and complex environments [29].

The group interference effect refers to the opinion deviation caused by the interactions and influences among decision-makers throughout the process of group decision-making [30]. The interference effects are often overlooked in traditional group decision models, which assume that each decision-maker evaluates information independently and makes a judgment. However, in reality, the opinions and choices of decision-makers are significantly influenced by the opinions and emotions of those ones around them. While, quantum probability theory can simulate this interaction, providing a more realistic way to describe and predict the decision-making behavior of human groups. This enables quantum models to effectively reveal paradoxes, sub-optimal choices, and irrational behaviors in human decision-making, particularly when describing group decision-making dynamics and considering complex psychological and social interactions among decision-makers [31,32]. On one hand, interference effects among decision-makers can be considered to simulate conflicts and ambiguities within groups, thus better-achieving group consensus preferences [33,34]. On the other hand, the calculation of the final probability is influenced by the quantum interference effect, thereby affecting the decision outcome [35–37]. In quantum theory, the interference value is usually expressed by the phase angle, but this angle is often uncertain. To measure the interference angle more accurately and better reflect the interaction between decision-makers [38,39], researchers have introduced belief entropy as an index to calculate probability uncertainty. This approach not only enhances the quality of decision-making but also guarantees the thoroughness and precision of the outcomes [29].

There are some challenges in the early decision-making research of meteorological disaster groups, especially when dealing with complex and uncertain decision-making environments. These challenges are mainly reflected in the following three aspects:

- 1) Interference effect among decision makers. Most of the existing research on group decision making is based on the traditional decision model, which does not fully consider the mutual interference effect among decision makers [1,2]. In the actual decision-making process, the opinions of decision makers often affect each other, especially in the face of high uncertainty environment, this mutual interference effect will have a significant impact on the group consensus. Most existing studies assume that the

Table 1

The comparison of the proposed method and previous decision models.

Models	Information	Decision makers' weights	Preference adjusting process	Attributes' weights	Interference term calculation
MAGDM [29]	Crisp number	-	No	Unknown	Deng entropy
MCGDM [36]	LTS	Unknown	No	Unknown	Phase Angle evolution
MCGDM [40]	I2TFL	-	No	Unknown	No Considered
GDM [15]	Crisp number	Known	Group consensus adjustment	Known	No Considered
GDM [41]	Crisp number	-	-	-	Similarity heuristic approach
GDM [34]	Crisp number	Known	Individual preference adjustment	Unknown	Similarity heuristic approach
The proposed model	LZN	Unknown	Group consensus adjustment	Unknown	Deng entropy

evaluations of decision makers are independent of each other, ignoring the complex interactions that exist in the actual situation.

- 2) Limitations of quantum decision models. Although quantum decision models can deal with uncertainty problems in theory, the existing literature rarely applies this framework in complex uncertain decision environments [29,30]. The main challenge is that it is difficult to obtain accurate quantitative data in fuzzy and uncertain environments, and existing models are mostly based on deterministic data derivation. However, existing studies generally assume that the interference angle is known, and it is calculated based on a specific angle, or only consider the direction of interference, ignoring the chaotic characteristics and variability of the system [32,33,36]. In addition, there are few researches on group consensus under the quantum framework.
- 3) The influence of uncertainty on attributes' weights is not considered. Traditional methods assume high weight certainty, or assign weight subjectively by decision makers, while ignoring the actual ambiguity and uncertainty in decision-making [15]. Moreover, in a complex uncertain environment, attributes' weights should be dynamically changed, and the influence of information changes in the decision-making process should be considered.

As shown in Table 1, most of the existing group decision-making models either do not consider the interference effect between groups, or assume that the weights of decision makers and attributes are known. In addition, research on group consensus processes within a quantum framework is limited. In the context of uncertainty related to meteorological disasters, LZNs is introduced to characterize the assessment information, the weights of decision makers are calculated through the group consensus process, and the uncertainty of information is captured by using Deng entropy. This approach more accurately reflects the interactions between decision makers and better matches reality.

To address these gaps, this paper attempts to answer the following research questions: How can LZNs be effectively used to deal with information uncertainty in meteorological disaster decision-making? In what ways can the quantum group consensus decision model enhance the coordination of conflicting viewpoints among decision makers? How to use maximum Deng entropy to accurately calculate attributes' weights to get better decision results? What are the advantages and practical implications of integrating quantum probability into Bayesian networks for evaluating contingency scenarios?

The motivation of this paper can be divided into the following three points: 1) Information uncertainty in the decision-making of coping with meteorological disasters. The uncertainty of information is a common challenge in the emergency decision-making process of meteorological disasters, and decision makers are often faced with fuzzy and complex environments. It is difficult for existing decision models to provide effective decision support when dealing with such highly uncertain information. 2) Improve the conflict coordination ability in group consensus. In the group decision-making process, especially when dealing with complex scenarios such as emergency management, there are usually significant differences in the views of decision makers. How to effectively coordinate these conflicts becomes the key to decision-making, and traditional decision-making models have limitations in coordinating conflicts. 3) Optimize the accuracy of decision weight calculation. The attribute's weights in the decision-making process is very important to the quality of the final decision result. In the face of complex and uncertain environment, the determination of attribute's weights often lacks sufficient accuracy.

The main contributions of this paper are as follows:

- 1) In this paper, a quantum group consensus method based on linguistic Z-numbers is proposed. The traditional multi-attribute group decision-making model usually assumes that the evaluation of decision makers is independent of each other, but in the real complex decision-making environment, the opinions and judgments of decision makers often affect each other. By introducing interference effects into the quantum framework, the model is able to more truly reflect the interaction between decision makers, and thus more accurately simulate the process of reaching group consensus in the face of differences of opinion and information asymmetry.
- 2) The method in this paper extends the quantum decision method to the fuzzy domain, and uses the LZNs to represent the information. Based on the traditional quantum decision model, this paper further applies it to fuzzy information processing and uses LZNs to represent uncertain information. In the context of dealing with uncertain and fuzzy information, LZNs provides a more flexible representation and expands the application scenarios of quantum decision models.
- 3) In this paper, the maximum Deng entropy model is used to calculate the attributes' weights. Deng entropy can deal with fuzzy and incomplete information better in the process of group decision making, especially when dealing with fuzzy and uncertain en-

vironment. By maximizing system uncertainty, Deng entropy can dynamically adjust attributes' weights to adapt to uncertainties brought about by changes in information over time and environment.

The structure of the paper is as follows: Section 2 reviews the LZNs, Deng entropy, group consensus measurement, and the framework of quantum probability theory. Section 3 introduces the multi-attribute quantum group consensus decision model (Z-MAQGCDM), which combines group consensus measures, Deng entropy, quantum probability, and Bayesian networks, and describes the decision steps in detail. In Section 4, the Z-MAQGCDM is illustrated through meteorological disaster emergency rescue cases, and the stability and dependability of the model have been confirmed through sensitivity and comparison analysis. In conclusion, Section 5 provides a concise overview of the primary discoveries and contributions made in this work, as well as proposes potential avenues for further research.

2. Preliminaries

This section aims to review key concepts and theories, laying a solid theoretical foundation for further discussion in the following chapters. Topics covered include linguistic Z-numbers, Deng entropy, group consensus processes, and quantum probability theory. Through an in-depth discussion of these theories, a comprehensive theoretical framework is constructed to provide the necessary theoretical support for the analysis and research in the subsequent chapters.

2.1. Linguistic scale function (LSF)

Definition 1 [4]. Let L be a linguistic set consisting of $2g$ elements, denoted as $L = \{l_0, l_1, \dots, l_{2g}\}$, where g is a positive integer. The LSF H is a mapping that assigns each element l_i in L to a corresponding element ζ_i .

$$H : l_i \rightarrow \zeta_i (i = 0, 1, \dots, 2g). \quad (1)$$

Where the symbol ζ_i reflects the expert's preference for using the linguistic term l_i and $0 \leq \zeta_0 < \zeta_1 < \dots < \zeta_{2g} \leq 1$. Possible functions of the LSF $H_o(l_i) (o = 1, 2, 3, 4)$ are as follows [42].

$$H_1(l_i) = \frac{i}{2g}, 0 \leq i \leq 2g. \quad (2)$$

$$H_2(l_i) = \left(\frac{i}{2g}\right)^g, 0 \leq i \leq 2g. \quad (3)$$

$$H_3(l_i) = \left(\frac{i}{2g}\right)^{\frac{1}{g}}, 0 \leq i \leq 2g. \quad (4)$$

$$H_4(l_i) = \begin{cases} \frac{b^g - b^{g-i}}{2b^g - 2}, 0 \leq i \leq g. \\ \frac{b^g + b^{i-g} - 2}{2b^g - 2}, g + 1 \leq i \leq 2g. \end{cases} \quad (5)$$

The value of b can be ascertained utilizing subjective methodologies. If assume that indication A has significantly more weight than indicator B , with an importance ratio of m , then researchers may express the relationship as $b^k = m$ (where k denotes the scale level) or $b = \sqrt[k]{m}$. The consensus among academics is that a significance ratio of $m = 9$ is the upper limit. At a scale level of 2, the value of a is nearly equivalent to the cube root of 9, which is about 2.08.

2.2. Z-numbers and linguistic Z-numbers

Decision-makers possess an exceptional capacity to make logical conclusions while having ambiguous, inaccurate, or missing information. To better describe decision information in real life, Zadeh introduced the concept of the Z-numbers.

Definition 2 [42]. Let X represent the domain of the argument. $L_1 = \{l_0, l_1, \dots, l_{2g}\}$ and $L_2 = \{l'_0, l'_1, \dots, l'_{2g'}\}$ be two finite and fully ordered sets of discrete linguistic terms. g and g' are non-negative integers. Furthermore, let $A_{\varphi(x)} \in L_1$ and $B_{\phi(x)} \in L_2$. The linguistic Z-set Z in X is a specific entity that has the following structure:

$$Z = \{(x, A_{\varphi(x)}, B_{\phi(x)}) | x \in X\}. \quad (6)$$

The variable Z is defined by two linguistic terms, $A_{\varphi(x)}$ and $B_{\phi(x)}$. $A_{\varphi(x)}$ restricts the range of possible values for the uncertain variable, while $B_{\phi(x)}$ measures its reliability. The sets of linguistic terms L_1 and L_2 are distinct and provide different preference information.

When the set X has just one element, the set of LZN simplifies to $(A_{\phi(x)}, B_{\phi(x)})$, which is denoted for convenience as $z = (A_{\varphi(x)}, B_{\phi(x)})$. Where $A_{\varphi(x)} \in L_1$ and $B_{\phi(x)} \in L_2$ are two linguistic terms.

Definition 3 [42]. Let $Z_x = (A_{\varphi(x)}, B_{\phi(x)})$ and $Z_{x'} = (A_{\varphi(x')}, B_{\phi(x')})$ represent two LZNs. f^* and g^* are LSFs derived from $H_o(\cdot) (o = 1, 2, 3, 4)$, and they pertain to the linguistic term sets A and B . The inverse functions of f^* and g^* are denoted by f^{*-1} and g^{*-1} . The symbols $\varphi(x) (x = 0, 1, \dots, 2g)$ and $\phi(x') (x' = 0, 1, \dots, 2g')$ correspond to the linguistic terms A_x and $B_{x'}$. The arithmetic operations for LZNs are defined as follows:

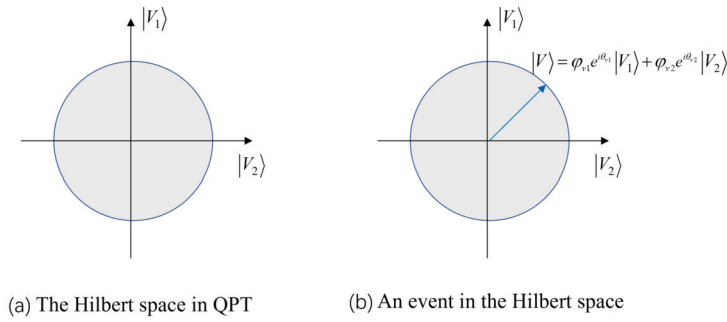


Fig. 1. Events in Hilbert space.

- 1) $Z_x \oplus Z_{x'} = (f^{*-1}(f^*(A_{\varphi(x)}) + f^*(A_{\varphi(x')})), g^{*-1}(\frac{f^*(A_{\varphi(x)})g^*(B_{\phi(x')}) + f^*(A_{\varphi(x')})g^*(B_{\phi(x)})}{f^*(A_{\varphi(x)}) + f^*(A_{\varphi(x')})}))$.
- 2) $Z_x \otimes Z_{x'} = (f^{*-1}(f^*(A_{\varphi(x)}) \times f^*(A_{\varphi(x')})), g^{*-1}(g^*(B_{\phi(x)}) \times g^*(B_{\phi(x')})))$.
- 3) $\lambda Z_x = (f^{*-1}(\lambda f^*(A_{\varphi(x)})), B_{\phi(x)}), \lambda \geq 0$.
- 4) $Z_x^\lambda = (f^{*-1}(f^*(A_{\varphi(x)})^\lambda), g^{*-1}(g^*(B_{\phi(x)})^\lambda)), \lambda \geq 0$.

Definition 4 [42]. The weighted Euclidean distance between $z_x = (A_{\varphi(x)}, B_{\phi(x)})$ and $z_{x'} = (A_{\varphi(x')}, B_{\phi(x')})$ is defined as follows:

$$d(z_x, z_{x'}) = \sqrt{\rho_1(f^*(A_{\varphi(x)}) - f^*(A_{\varphi(x')}))^2 + \rho_2(g^*(B_{\phi(x)}) - g^*(B_{\phi(x')}))^2}. \quad (7)$$

Where $\rho_1 \in [0, 1]$ and $\rho_2 \in [0, 1]$ are the parameters reflecting the importance of A and B in the LZNs, and $\rho_1 + \rho_2 = 1$. If component A is deemed to have greater significance than part B , then $\rho_1 \geq \rho_2$. Specifically, when both ρ_1 and ρ_2 are equal to 0.5, the importance of A and B is equal.

2.3. Deng entropy

Definition 5 [28]. Given a mass function m defined on the Frame of Discernment (FOD) $\Theta = \{\theta_1, \theta_2, \dots, \theta_N\}$, Deng entropy is defined as follows:

$$E_D(m) = - \sum_{Y \in 2^\Theta} m(Y) \log_2 \left[\frac{m(Y)}{2^{|Y|} - 1} \right]. \quad (8)$$

Where Y is the focus element.

2.4. Quantum probability theory

All possible occurrences are seen as subspaces inside Hilbert spaces in quantum probability theory, in contrast to classical probability theory.

Definition 6 [36]. Hilbert space is a mathematical concept that expands upon Euclidean space by allowing for infinite dimensions. Consider a real linear space H . If x^0 and y^0 are any two vectors in H that correspond to a real number, denoted as (x^0, y^0) , then the following conditions are satisfied:

- 1) For any two vectors x^0 and y^0 in H , there is $(x^0, y^0) = (y^0, x^0)$.
- 2) For any three vectors x^0, y^0, z^0 and the crisp numbers α^0, β in H , there is $(\alpha^0 x^0 + \beta y^0, z^0) = \alpha^0 (x^0, z^0) + \beta (y^0, z^0)$.
- 3) For every vector x^0 in H , there is $(x^0, x^0) \geq 0$, and the necessary and sufficient condition for $(x^0, x^0) = 0$ is $x^0 = 0$. Then (x^0, y^0) is defined to be an inner product on H , and H is defined the inner product space.

Constitutes a complex vector space spanned by a set of orthogonal basis vectors $V = \{|V_1\rangle, \dots, |V_n\rangle\}$. In Hilbert space, events are defined using quantum superposition states $|D\rangle$, which may be expressed as linear combinations of orthogonal basis vectors, as seen in Fig. 1.

$$\begin{aligned} |D\rangle &= \langle V_1 | D \rangle |V_1\rangle + \langle V_2 | D \rangle |V_2\rangle + \dots + \langle V_n | D \rangle |V_n\rangle \\ &= \psi_1 |V_1\rangle + \psi_2 |V_2\rangle + \dots + \psi_n |V_n\rangle \\ &= \varphi_1 e^{i\theta_1} |V_1\rangle + \varphi_2 e^{i\theta_2} |V_2\rangle + \dots + \varphi_n e^{i\theta_n} |V_n\rangle. \end{aligned} \quad (9)$$

Furthermore, using the complex $\psi_k, \varphi_k e^{i\theta_k} (k = 1, 2, \dots, n)$ to describe the probability amplitude of state $V_k (k = 1, 2, \dots, n)$, where $(\varphi_k)^2 \in [0, 1]$, $\sum_{k=1}^n (\varphi_k)^2 = 1$. The expression $e^{i\theta_k}$ is used to denote the phase of this complex wave function. Based on Born's rule, the square of the wave function, $(\varphi_k)^2$ is exactly proportional to the chance of seeing that state.

$$\begin{aligned} P(V_k) &= |\varphi_k e^{i\theta_k}|^2 \\ &= (\varphi_k e^{i\theta_k}) \times (\varphi_k e^{i\theta_k})^* \\ &= \varphi_k e^{i\theta_k} \times \varphi_k e^{-i\theta_k} = (\varphi_k)^2, k = 1, 2, \dots, n. \end{aligned} \quad (10)$$

3. Multi-attribute group consensus model with linguistic Z-numbers considering interference effects

The primary objective of the MAGDM problem is to obtain a solution that is based on consensus, considering various alternatives and their corresponding attributes. The essential components of a MAGDM problem are as follows.

- 1) A group of decision makers $D = \{d_1, d_2, \dots, d_s\} (s \geq 2)$, the set of decision makers' weights is $\Lambda = \{\lambda_1, \lambda_2, \dots, \lambda_s\}$, satisfying the condition $\lambda_k \geq 0 (k = 1, 2, \dots, s)$ and $\sum_{k=1}^s \lambda_k = 1$.
- 2) A limited set of alternative alternatives, called $A' = \{a_1, a_2, \dots, a_m\} (m \geq 2)$, are potential solutions to the MAGDM problem.
- 3) An attributes set $C = \{c_1, c_2, \dots, c_n\} (n \geq 2)$, attribute weights sets $W = \{w_1, w_2, \dots, w_n\}$, satisfy the condition $w_j \geq 0 (j = 1, 2, \dots, n)$ and $\sum_{j=1}^n w_j = 1$.

3.1. Collect evaluation information for decision makers

1) Evaluation information collection. Each decision-maker $d_k (k = 1, 2, \dots, s)$ evaluates the alternative $a_i (i = 1, 2, \dots, m)$ about attribute $c_j (j = 1, 2, \dots, n)$ by the linguistic Z-numbers $z_{ij}^k = (A_{ij}^k, B_{ij}^k)$. The evaluation information matrix is denoted as $d_k (k = 1, 2, \dots, s)$.

$$Z^k = \begin{pmatrix} z_{11}^k & z_{12}^k & \dots & z_{1n}^k \\ z_{21}^k & z_{22}^k & \dots & z_{2n}^k \\ \vdots & \vdots & \ddots & \vdots \\ z_{m1}^k & z_{m2}^k & \dots & z_{mn}^k \end{pmatrix}, k = 1, 2, \dots, s. \quad (11)$$

On this basis, the evaluation information matrix $d_k (k = 1, 2, \dots, s)$ can be transformed into the linguistic function matrix $d_k (k = 1, 2, \dots, s)$.

2) To prevent information loss, this paper uses relative tightness to describe the relative distance between decision-making units (DMUs) and ideal and non-ideal solutions [42]. By reflecting the proximity of $\hat{z}_{ij}^k$ with respect to the ideal and non-ideal solutions, and converting the linguistic Z-numbers to precise numbers, we obtain the clear number decision matrix $d_k (k = 1, 2, \dots, s)$.

$$\hat{z}_{ij}^k = \frac{d(\hat{z}_{ij}^k, \hat{z}^-)^2}{d(\hat{z}_{ij}^k, \hat{z}^+)^2 + d(\hat{z}_{ij}^k, \hat{z}^-)^2}, i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s. \quad (12)$$

Where, $z_{ij}^k \in [0, 1]$ represents the initial information assessment value of the decision maker d_k on the attribute $c_j \in C$ for the alternative $a_i \in A'$. Then, $\hat{z}^+$ and $\hat{z}^-$ correspond to the clear linguistic Z-numbers of the best linguistic Z-numbers $\hat{z}^+ = (A_{2g}, B_{2g'})$ and the worst linguistic Z-numbers $\hat{z}^- = (A_0, B_0)$ for the clear linguistic Z-numbers.

3.2. Decision makers' weights determination based on group consensus adjustment

In the process of promoting consensus at the lowest cost, decision-makers make acceptance or rejection decisions based on feedback suggestions based on their cognitive abilities and the adjustment compensation they receive from top decision-makers. The objective of this procedure is to minimize the cost while adjusting the degree of consensus. It tries to optimize the cost-benefit ratio in order to guarantee efficient and economical decision-making.

3.2.1. Group consensus measure

Consensus measuring is a repetitive process of modifying preferences with the goal of enhancing the degree of agreement within a group [16]. Firstly, the top decision maker gives the initial set of weights for each decision maker $\Lambda = (\lambda_1^0, \lambda_2^0, \dots, \lambda_s^0)$, and the Ordered Weighted Averaging (OWA) operator is employed to combine the group information assessment matrix $Z^o = (z_{ij}^o)_{m \times n}$.

$$z_{ij}^o = \sum_{k=1}^s \lambda_k^0 \hat{z}_{ij}^k, i = 1, 2, \dots, m, j = 1, 2, \dots, n. \quad (13)$$

The Manhattan distance [15] is employed to quantify the distance between individual assessment information and group assessment information, and the individual consensus level is computed as follows.

$$IC(d_k) = 1 - \frac{\sum_{i=1}^m \sum_{j=1}^n |z_{ij}^k - z_{ij}^o|}{mn}, k = 1, 2, \dots, s. \quad (14)$$

The group consensus level is determined by combining the individual consensus level using the OWA operator.

$$GC = \sum_{k=1}^s \lambda_k^0 IC(d_k). \quad (15)$$

The degree of group agreement increases as the value of GC increases within the range of 0 to 1. The group consensus threshold, denoted by μ , also falls within the range of 0 to 1. If the value of GC is less than μ , the existing degree of group consensus is deemed inappropriate or unsatisfactory, and improvements are required via feedback. Otherwise, the current group information evaluation matrix is considered the final matrix.

3.2.2. Group consensus feedback mechanism

At this stage, the top decision maker improves the group consensus level by providing feedback suggestions. The threshold of group consensus is set as $\mu \in [0, 1]$. If the group consensus level $GC < \mu$, the consensus level of the current group is identified as unacceptable or unsatisfactory. Therefore, it is necessary to identify the decision makers d_k ($k = 1, 2, \dots, s$) who have not reached a satisfactory consensus level and provide them with feedback suggestions for adjusting their personal preferences to improve the consensus level through adjustments. Here, $\bar{Z}^k = (\bar{z}_{ij}^k)_{m \times n}$ ($k = 1, 2, \dots, s$) is the feedback advice provided to the decision maker d_k ($k = 1, 2, \dots, s$). The feedback information matrix $\bar{z}_{ij}^k$ ($i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$) is defined as:

$$\bar{z}_{ij}^k \in [\min d(\hat{z}_{ij}^k, z_{ij}^o), \max d(\hat{z}_{ij}^k, z_{ij}^o)], i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s. \quad (16)$$

Each decision maker adjusts the evaluation information based on the feedback information, generating the decision matrix $\hat{Z}^{k(1)} = (\hat{z}_{ij}^{k(1)})_{m \times n}$ ($k = 1, 2, \dots, s$) after the first round of adjustment. Then, Eqs. (13)-(15) are applied to compute the first-round adjusted group consensus level and compare it with the preset group consensus threshold μ . For example, if $GC^{(1)} < \mu$ ($GC^{(1)}$ represents the group consensus level of the first round of adjustment), it indicates that further adjustments are needed to achieve a higher consensus level. Otherwise, the current matrix $\bar{Z}^{k(1)} = (\bar{z}_{ij}^{k(1)})_{m \times n}$ ($k = 1, 2, \dots, s$) can be considered the final decision matrix.

3.2.3. Group consensus optimization model based on minimum compensation cost

The top decision maker will provide adjustment compensation $m_{ij}^{k(t)}$ ($i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s, t = 1, 2, \dots, h$) to individual decision makers d_k ($k = 1, 2, \dots, s$) to guide them to adjust their opinions. Here, $m_{ij}^{k(t)}$ ($i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s, t = 1, 2, \dots, h$) represents the adjustment compensation given to the decision maker by the top decision maker for the attribute c_j ($j = 1, 2, \dots, n$) of alternative a_i ($i = 1, 2, \dots, m$) in the t -th ($0 \leq t \leq h$) round of adjustment.

Suppose that the decision makers d_k ($k = 1, 2, \dots, s$) have their confidence thresholds [43] b_j^k ($j = 1, 2, \dots, n, k = 1, 2, \dots, s$) for each attribute c_j ($j = 1, 2, \dots, n$) under different decision conditions. If the top decision maker wishes to foster consensus when a decision maker's adjusted opinion exceeds their confidence threshold, measures must be taken to change the decision maker's compromise interval. These solutions may involve augmenting remuneration for the choice maker or modifying the decision conditions, both of which necessitate supplementary exertion from the principal decision maker. Based on this, the following adjustment cost function is as follows.

$$m_{ij}^{k(t)} = \begin{cases} \eta_{ij}^{k(t)}, & |z_{ij}^k - \bar{z}_{ij}^k| \leq b_j^k \\ \eta_{ij}^{k(t)} + \delta_{ij}^{k(t)}, & b_j^k \leq |z_{ij}^k - \bar{z}_{ij}^k| \end{cases} \quad (17)$$

$i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s, t = 1, 2, \dots, h.$

Thus, the model for optimizing group consensus, which is based on minimizing compensation cost, the model as follows:

$$\begin{aligned} \min & \frac{1}{smn} \sum_{k=1}^s \sum_{i=1}^m \sum_{j=1}^n m_{ij}^{k(t)} |\bar{z}_{ij}^k - z_{ij}^k| \\ & \begin{cases} \bar{z}_{ij}^k \in [\min d(\hat{z}_{ij}^k, z_{ij}^o), \max d(\hat{z}_{ij}^k, z_{ij}^o)] \\ z_{ij}^o = \sum_{k=1}^s \lambda_k^0 \hat{z}_{ij}^k \\ \bar{z}_{ij}^0 = \sum_{k=1}^s \lambda_k^0 \bar{z}_{ij}^k \\ m_{ij}^{k(t)} = \begin{cases} \eta_{ij}^{k(t)}, & |\hat{z}_{ij}^k - \bar{z}_{ij}^k| \leq b_j^k \\ \eta_{ij}^{k(t)} + \delta_{ij}^{k(t)}, & b_j^k \leq |\hat{z}_{ij}^k - \bar{z}_{ij}^k| \end{cases} \\ \sum_{k=1}^s \lambda_k^0 \left(1 - \frac{\sum_{i=1}^m \sum_{j=1}^n |\bar{z}_{ij}^k - \bar{z}_{ij}^0|}{mn} \right) = GC^* \\ \eta_{ij}^{k(t)} \geq 0, \delta_{ij}^{k(t)} \geq 0, \bar{z}_{ij}^k \in [0, 1], GC^* \in [0, 1] \\ i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s, t = 1, 2, \dots, h \end{cases} \end{aligned} \quad (18)$$

Theorem 1. Model (18) has an optimal solution.

Proof. Firstly, the objective function is a triple additive absolute value function, which is continuous and non-negative. Moreover, due to the property of the absolute value function, the objective function is bounded lower in the feasible solution space, which has one of the basic conditions for the existence of an optimal solution. Next, the constraints guarantee knowledge of the closed set and boundedness of the space. Firstly, the value of the decision variable $\bar{z}_{ij}^k$ ($i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$) is restricted in a closed interval to ensure the boundedness of the space. Secondly, the linear combination constraints of z_{ij}^o ($i = 1, 2, \dots, m, j = 1, 2, \dots, n$) and $\bar{z}_{ij}^0$ ($i = 1, 2, \dots, m, j = 1, 2, \dots, n$) ensure their linear continuity. The value rule of $m_{ij}^{k(t)}$ ($i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s, t = 1, 2, \dots, h$) is definite and non-negative, which does not affect the existence of the solution. The constraint of group agreement level indicates the calculation method of group consensus, and ensures the boundedness and continuity of the value range. Therefore, the entire feasible solution space is closed and bounded. According to Weierstrass theorem, the continuous function must have an optimal solution on the compact feasible domain [44]. Therefore, it can be concluded that Model (18) has an optimal solution. $\square$

3.2.4. Decision-maker's weight calculation

This paper proposes the utilization of an exponential function for determining the weight of decision makers. Specifically, the weights are computed based on the disparity between the decision matrix and the group aggregation matrix. As the distance increases, indicating a greater deviation between the matrices, the weight assigned to the decision-maker decreases accordingly.

$$\lambda_k = \frac{e^{-\sum_{i=1}^m \sum_{j=1}^n d(\hat{z}_{ij}^{k(h)}, z_{ij}^{o(h)})}}{\sum_{k=1}^s e^{-\sum_{i=1}^m \sum_{j=1}^n d(\hat{z}_{ij}^{k(h)}, z_{ij}^{o(h)})}}, k = 1, \dots, s. \quad (19)$$

Where $Z^{o(h)} = (z_{ij}^{o(h)})_{m \times n}$ represents the final decision matrix of each decision maker $\hat{Z}^{k(h)} = (\hat{z}_{ij}^{k(h)})_{m \times n}$ ($k = 1, 2, \dots, s$) applying the OWA operator to the aggregated collective choice matrix.

Theorem 1. Let λ_k ($k = 1, 2, \dots, s$) be the weight assigned to the k -th decision-maker, Satisfy the following properties:

- 1) *Sensitivity:* λ_k ($k = 1, 2, \dots, s$) decreases exponentially as the distance $\sum_{i=1}^m \sum_{j=1}^n d(\hat{z}_{ij}^{k(h)}, z_{ij}^{o(h)})$ increases, ensuring that larger deviations from the group consensus are penalized more heavily.
- 2) *Normalization:* The weights λ_k ($k = 1, 2, \dots, s$) are normalized such that $\sum_{k=1}^s \lambda_k = 1$.
- 3) *Exclusivity:* The exponential function naturally excludes the occurrence of non-zero and negative weights. In contrast, other functional forms (e.g., linear or polynomial functions) may introduce undesirable results such as negative weights or biased weight distributions.

Proof. 1) Sensitivity. To show that λ_k ($k = 1, 2, \dots, s$) decreases exponentially with increasing distance, consider the exponent $-\sum_{i=1}^m \sum_{j=1}^n d(\hat{z}_{ij}^{k(h)}, z_{ij}^{o(h)})$. As the distance $\sum_{i=1}^m \sum_{j=1}^n d(\hat{z}_{ij}^{k(h)}, z_{ij}^{o(h)})$ increases, the exponent becomes more negative, making $e^{-\sum_{i=1}^m \sum_{j=1}^n d(\hat{z}_{ij}^{k(h)}, z_{ij}^{o(h)})}$ smaller. This demonstrates that larger deviations result in exponentially smaller weights. $\square$

2) Normalization. The normalization property follows directly from the definition of $\lambda_k (k = 1, 2, \dots, s)$. By summing over all decision-makers, the result is obtained:

$$\sum_{k=1}^s \lambda_k = \sum_{k=1}^s \frac{e^{-\sum_{i=1}^m \sum_{j=1}^n d(z_{ij}^{k(h)}, z_{ij}^{o(h)})}}{\sum_{k=1}^s e^{-\sum_{i=1}^m \sum_{j=1}^n d(z_{ij}^{k(h)}, z_{ij}^{o(h)})}} = 1.$$

This confirms that the weights $\lambda_k (k = 1, 2, \dots, s)$ are normalized. $\square$

3) Exclusivity. The exponential function form ensures non-negativity and zero-bound constraints for the weights. For example, when the distance $\sum_{i=1}^m \sum_{j=1}^n d(z_{ij}^{k(h)}, z_{ij}^{o(h)})$ is large, the weight $\lambda_k (k = 1, 2, \dots, s)$ quickly approaches zero but never becomes negative. In contrast, linear or polynomial functions might yield negative weights or inappropriate weight distributions when dealing with large distance variations. $\square$

3.3. Attributes' weights determination based on Deng entropy

In multi-attribute decision analysis, Deng entropy demonstrates significant advantages due to its high accuracy in measuring uncertainty. Based on the concept of entropy, Deng entropy aims to measure the disordered state of system information and extract and utilize information between attributes by maximizing entropy. Therefore, according to the principle of maximum entropy, the weight allocation that achieves maximum entropy under known information constraints is considered the most ideal solution. This method deduces weights directly from the distribution of attribute data, ensuring the full use of information and the objectivity of weight allocation in the decision-making process.

By combining the theory of entropy with the structure of the LZN, a reasonable weight determination model can be constructed. In this model, the entropy of the weights should be maximized while meeting several key constraints: 1) The whole weight of all attributes must be equal to 1; 2) According to the relative relationship between matrix elements, the weight of each attribute is maintained within a reasonable range, with the weight value interval for each attribute determined by the maximum and minimum values of each column in the decision matrix; 3) Control the variance fluctuation of attribute weights. Therefore, after obtaining the final decision matrix $\hat{Z}^{k(t)} = (z_{ij}^{k(t)})_{m \times n} (k = 1, 2, \dots, s)$ following the adjustment round t in Section 3.2 of this paper, the new decision-maker weights are applied to aggregate the decision matrix, resulting in the final collective decision matrix $Z^{o(h)}$. The maximum entropy model of attribute weights is as follows:

$$\begin{aligned} \max F = & - \sum_{j=1}^n w_j \log_2 \frac{w_j}{2^n - 1} \\ \text{s.t.} \left\{ \begin{array}{l} \sum_{j=1}^n w_j = 1 \\ w_j \in \left(\min_i \frac{z_{ij}^{o(h)}}{\sum_{i=1}^m z_{ij}^{o(h)}}, \max_i \frac{z_{ij}^{o(h)}}{\sum_{i=1}^m z_{ij}^{o(h)}} \right) \\ \frac{1}{n} \sum_{j=1}^n \left(w_j - \frac{1}{n} \right)^2 \in (\min_i D(j), \max_i D(j)) \\ D(j) = \frac{1}{m} \sum_{i=1}^m \left(\frac{z_{ij}^{o(h)}}{\sum_{i=1}^m z_{ij}^{o(h)}} - \frac{1}{m} \right) \\ i = 1, 2, \dots, m, j = 1, 2, \dots, n \end{array} \right. \end{aligned} \quad (20)$$

Theorem 2. The Model (20) has an optimal solution.

Proof. Firstly, the objective function F is a concave function with respect to w_j , and this is because it is possible to write the objective function as the negative of a convex function, which is $F = -h(w_1, w_2, \dots, w_n)$, where $h(w_1, w_2, \dots, w_n)$ is a convex function. By the nature of concave functions, any local minimum of a concave function over its domain of definition is also a global minimum. Secondly, constraint (1) defines a convex set, i.e., the unit hyperplane, constraint (2) defines the feasible domain of w_j , and constraint (3) is a range constraint, which together forms a closed convex feasible domain. Combining the concavity of the objective function and the convexity of the constraints, the KKT conditions from the theory of convex optimization can be applied [45]. These conditions provide necessary and sufficient conditions for the optimality of the convex optimization problem. Therefore, according to the KKT conditions, it can be concluded that there exists an optimal solution to the Model (20). $\square$

3.4. Quantum probability calculations considering interference effects

In the context of meteorological disaster relief, two layers of the Bayesian network describe the Z-MAQGCDM process. This process combines quantum decision-making theory with the Bayesian decision framework to provide innovative tools for the planning and

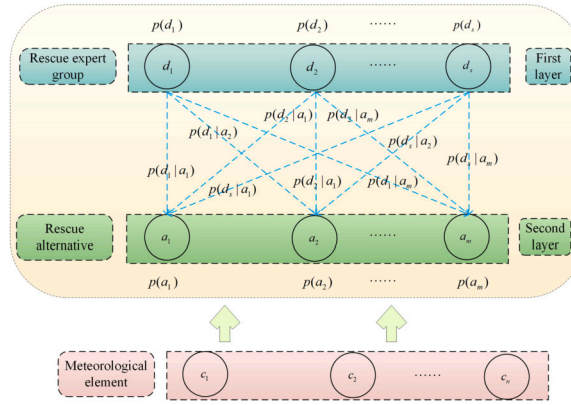


Fig. 2. Bayesian network structure of Z-MAQGCDM model in meteorological disaster.

implementation of emergency response. The initial layer of the quantum-like Bayesian network represents the decision group, whose views are influenced and disturbed by external influences. The specific decision structure is shown in Fig. 2.

1) The weight of the decision maker constitutes the first layer probability of the quantum-like Bayesian network.

$$P(d_k) = \lambda_k, k = 1, 2, \dots, s. \quad (21)$$

2) The conditional probability of the alternative a_i can be calculated as follows:

$$P(a_i|d_k) = \frac{\sum_{j=1}^n w_j z_{ij}^{k(h)}}{\sum_{i=1}^m \sum_{j=1}^n w_j z_{ij}^{k(h)}}, i = 1, 2, \dots, m, k = 1, 2, \dots, s. \quad (22)$$

3) To determine the cosine of the interference phase angle of the decision maker, follow these particular steps:

Firstly, the decision maker's preference matrix for alternatives $Q = (q_{ik})_{m \times s}$ is obtained based on the Bayesian network. q_{ik} ($i = 1, 2, \dots, m, k = 1, 2, \dots, s$) represents the decision maker d_k ($k = 1, 2, \dots, s$) for the alternatives a_i ($i = 1, 2, \dots, m$) preferences and satisfy $\sum_{i=1}^m \sum_{k=1}^s q_{ik} = 1$, calculated as follows:

$$q_{ik} = P(d_k)P(a_i|d_k), i = 1, 2, \dots, m, k = 1, 2, \dots, s. \quad (23)$$

On this basis, the quantum preference matrix $Q' = (q'_{ik} e^{i\theta_{ik}})_{m \times s}$ is computed, and the probability is determined by squaring the length of the projection.

$$q'_{ik} = \sqrt{q_{ik}}, i = 1, 2, \dots, m, k = 1, 2, \dots, s. \quad (24)$$

Secondly, a connection matrix consisting of connection vectors $\vec{\gamma}_{i,e}$ is constructed by combining pairs of the same choices under different decision-makers [29].

$$\begin{aligned} \vec{\gamma}_{i,1} &= \begin{bmatrix} q_{i1} \\ q_{i2} \end{bmatrix}, \vec{\gamma}_{i,2} = \begin{bmatrix} q_{i1} \\ q_{i3} \end{bmatrix}, \dots, \vec{\gamma}_{i,n-1} = \begin{bmatrix} q_{i1} \\ q_{im} \end{bmatrix}, \\ &\dots \\ \vec{\gamma}_{i, \frac{n(n-1)}{2}} &= \begin{bmatrix} q_{in-1} \\ q_{in} \end{bmatrix}, i = 1, 2, \dots, m. \end{aligned} \quad (25)$$

Where e is the number reassigned to the association between decision-makers. For example, when $e = 1$, $\vec{\gamma}_{i,1}$ denotes the connection vector between the first decision maker and the second decision maker.

Then, the confidence distance is calculated as follows:

$$M_{ie} = \left| v_{ie} + \frac{|v_{ie} - \eta| - |\vartheta_{ie} - \eta|}{|v_{ie} - \eta| + |\vartheta_{ie} - \eta|} \right|, i = 1, 2, \dots, m, e = 1, 2, \dots, \frac{s(s-1)}{2}. \quad (26)$$

If $|v_{ie} - \eta| < |\vartheta_{ie} - \eta|$, then the positions of v_{ie} and ϑ_{ie} should be interchanged and $\eta = \frac{1}{ms}$. On this basis, Deng entropy is calculated:

$$D_{ie}(M_{ie}) = -M_{ie} \log_2 \frac{M_{ie}}{2^m - 1}, i = 1, 2, \dots, m, e = 1, 2, \dots, \frac{s(s-1)}{2}. \quad (27)$$

In order to ensure that the values of the Deng entropy fall within the same range as the values of $\cos(\theta_{d_k} - \theta_{d_{\hat{k}}})$, which is between $[-1, 1]$, Deng entropy is standardized.

$$E_{ie}(M_{ie}) = 2 \frac{D_{ie}(M_{ie}) - \min\{D_{ie}(M_{ie})\}}{\max\{D_{ie}(M_{ie})\} - \min\{D_{ie}(M_{ie})\}} - 1, i = 1, 2, \dots, m, e = 1, 2, \dots, \frac{s(s-1)}{2}. \quad (28)$$

The cosine of the phase difference is determined by $E_{ie}(M_{ie})$.

$$\cos(\theta_{ik} - \theta_{i\hat{k}}) = E_{ie}(M_{ie}), i = 1, 2, \dots, m, e = 1, 2, \dots, \frac{s(s-1)}{2}. \quad (29)$$

Then the interference item can be expressed as:

$$\begin{aligned} interference_i &= 2 \sum_{k=1}^{s-1} \sum_{\hat{k}=k+1}^s \varphi_{d_k} \varphi_{a_i|d_k} \varphi_{d_{\hat{k}}} \varphi_{a_i|d_{\hat{k}}} \cos(\theta_{ik} - \theta_{i\hat{k}}) \\ &= 2 \sum_{k=1}^{s-1} \sum_{\hat{k}=k+1}^s \varphi_{d_k} \varphi_{a_i|d_k} \varphi_{d_{\hat{k}}} \varphi_{a_i|d_{\hat{k}}} E_{ie}(M_{ie}), i = 1, 2, \dots, m. \end{aligned} \quad (30)$$

Finally, the probability of each alternative is calculated:

$$\begin{aligned} P(a_i) &= \sigma \left| \sum_{k=1}^s \varphi_{d_k} e^{i\theta_{d_k}} \varphi_{a_i|d_k} e^{i\theta_{a_i|d_k}} \right|^2 \\ &= \sigma \left[\sum_{k=1}^s \left| \varphi_{d_k} \varphi_{a_i|d_k} \right|^2 + 2 \sum_{k=1}^{s-1} \sum_{\hat{k}=k+1}^s \varphi_{d_k} \varphi_{a_i|d_k} \varphi_{d_{\hat{k}}} \varphi_{a_i|d_{\hat{k}}} \cos(\theta_{ik} - \theta_{i\hat{k}}) \right] \\ &= \sigma \left[\sum_{k=1}^s \left| \varphi_{d_k} \varphi_{a_i|d_k} \right|^2 + interference_i \right], i = 1, 2, \dots, m. \end{aligned} \quad (31)$$

Where,

$$\sigma = \frac{1}{\sum_{i=1}^m \left[\sum_{k=1}^s \left| q'_{ik} \right|^2 + 2 \sum_{k=1}^{s-1} \sum_{\hat{k}=k+1}^s q'_{ik} q'_{i\hat{k}} \cos(\theta_{ik} - \theta_{i\hat{k}}) \right]}. \quad (32)$$

3.5. Decision steps

Z-MAQGCDM model considers the interference effect under the quantum framework. The model construction includes the following steps:

Step 1: Collect the evaluation information. Each decision maker $d_k (k = 1, 2, \dots, s)$ evaluates the attributes $c_j (j = 1, 2, \dots, n)$ in the alternatives $a_i (i = 1, 2, \dots, m)$ and obtains the initial evaluation information matrix $Z^k = (z_{ij}^k)_{m \times n} (k = 1, 2, \dots, s)$.

Step 2: Transform the initial information evaluation matrix into a decision matrix. According to the linguistic scale function, the evaluation information matrix is transformed into the linguistic scale function matrix $\tilde{Z}^k = (\tilde{z}_{ij}^k)_{m \times n} (k = 1, 2, \dots, s)$, and the decision matrix $\hat{Z}^k = (\hat{z}_{ij}^k)_{m \times n} (k = 1, 2, \dots, s)$ is transformed according to Eq. (12).

Step 3: Determining the weight of decision makers determined through group consensus. The initial group consensus level is calculated according to Eqs. (13)-(15). If this level is higher than the predetermined group consensus threshold, the final decision matrix is directly obtained. Otherwise, the top decision maker identifies the discrepancy and provides a feedback matrix to each decision maker according to Model (18) to adjust the group consensus until the consensus level exceeds the agreed threshold. The weight of each decision-maker $\lambda_k (k = 1, 2, \dots, s)$ is then calculated using Eq. (19).

Step 4: Determine the attribute weights with the combination of Deng entropy. The maximum entropy Model (20) based on Deng entropy is constructed to solve the optimal attributes' weights.

Step 5: Determine the quantum-like Bayesian network. The weight of the decision-maker $\lambda_k (k = 1, 2, \dots, s)$ is determined as the node probability $P(d_k) (k = 1, 2, \dots, s)$ in the first layer of the BN. The conditional probability $P(a_i|d_k) (i = 1, 2, \dots, m, k = 1, 2, \dots, s)$ in the BN is obtained from Eq. (22).

Step 6: Interference value calculation based on Deng entropy. The confidence distance is calculated using Eq. (26), the Deng entropy is calculated and normalized according to Eqs. (27)-(28), and the interference value is calculated using Eqs. (29)-(30).

Step 7: Quantum interference effects are taken into account in the alternative assessment. Eq. (31) is used to compute the quantum probability $P(a_i) (i = 1, 2, \dots, m)$ for each alternative, and the alternatives are arranged in ascending order based on their quantum probability values.

Due to time limitations during the consensus process, Algorithm 1 is often restricted to a maximum of five iterations [46]. Consequently, the linear programming model of Algorithm 1 is the critical component in addressing the computational complexity of the algorithm. In this paper, the gradient-based nonlinear optimization method is used to solve Model (18), utilizing the sequential quadratic programming (SQP) algorithm. The optimal solution is sought by calculating the gradient of the objective function and the Jacobian matrix of constraints. The specific process is detailed in Algorithm 1.

Algorithm 1 Z-MAQGCDM Model.

Input: Decision set $D = \{d_1, d_2, \dots, d_t\}$, alternative set $A' = \{a_1, a_2, \dots, a_m\}$, decision-maker weight set $\Lambda = \{\lambda_1, \lambda_2, \dots, \lambda_s\}$, attribute set $C = \{c_1, c_2, \dots, c_n\}$, attribute weight set $W = \{w_1, w_2, \dots, w_n\}$, confidence matrix $B^k = (b_1^k, b_2^k, \dots, b_n^k)$, linguistic size function $f(\cdot)$ and $g(\cdot)$, linguistic set $A = \{A_0, A_1, \dots, A_{2g}\}$ and $B = \{B_0, B_1, \dots, B_{2g'}\}$.

Output: The ranking results of alternatives.

Begin

Step 1: Collect evaluation information. The initial information evaluation matrix $Z^k = (z_{ij}^k)_{m \times n}$ ($k = 1, 2, \dots, s$) for each decision maker is obtained.

Step 2: The initial information evaluation matrix is transformed into a decision matrix. Application $f(\cdot)$ and $g(\cdot)$ for the linguistic function matrix scale $\tilde{Z}^k = (\tilde{z}_{ij}^k)_{m \times n}$ ($k = 1, 2, \dots, s$). According to Eq. (12), the decision matrix is given by $\hat{Z}^k = (\hat{z}_{ij}^k)_{m \times n}$ ($k = 1, 2, \dots, s$).

Step 3: Decision Maker Weight Determination Based on Group Consensus. According to Eqs. (13)–(15), the initial group consensus level $GC^{(0)}$ is calculated. If this level is higher than the predetermined group consensus threshold μ , the final decision matrix is directly obtained. Otherwise, Model (18) is solved to provide the feedback matrix $\tilde{Z}^{k(t)} = (\tilde{z}_{ij}^{k(t)})_{m \times n}$ ($k = 1, 2, \dots, s, t = 0, 1, \dots, h$) for each decision maker for group consensus adjustment. This process continues until the group consensus level is higher than the agreed-upon threshold. Finally, the weight of each decision maker λ_k ($k = 1, 2, \dots, s$) is calculated using Eq. (19).

Step 4: The attribute weights are calculated using Deng entropy, and the optimum attribute weights are found by solving the maximum Deng entropy model (20).

Step 5: Based on the QLBN. The probability of the first node is decision makers' weights, Eq. (22) is applied to obtain the conditional probability in Bayesian networks.

Step 6: Interference value calculation is based on Deng entropy. Eq. (26) is used to calculate confidence distance $M_{ie}(i = 1, 2, \dots, m, e = 1, 2, \dots, \frac{s(s-1)}{2})$. According to Eqs. (27)–(28), Deng entropy $D_{ie}(M_{ie})(i = 1, 2, \dots, m, e = 1, 2, \dots, \frac{s(s-1)}{2})$ and Deng entropy value standardization $E_{ie}(M_{ie})(i = 1, 2, \dots, m, e = 1, 2, \dots, \frac{s(s-1)}{2})$ are computed. Finally, Eq. (29) is used to calculate phase cosine $\cos(\theta_{d_i} - \theta_{d_j})$. According to Eq. (30), the interference item $interference_i(i = 1, 2, \dots, m)$ is calculated.

Step 7: Eq. (31) is applied to calculate the quantum probability of each alternative.

End

In order to further emphasize the mathematical formula and its logical relationship in this paper, Fig. 3 is used to visualize the description.

4. Case analysis

This section employs a meteorological disaster reaction to assess the practicality of the proposed Z-MAQGCDM. The case study provides a thorough examination that showcases the efficacy and superiority of the Z-MAQGCDM model in managing the interaction between decision-makers and the relationship between attributes.

4.1. Background

In dealing with emergency rescue operations during meteorological disasters, formulating an effective rescue plan is crucial. This paper uses Zhuozhou and its surrounding areas in July 2023 as a case study. The region experienced abnormal rainfall events, resulting in disasters such as urban waterlogging, flash floods, mudslides, and building collapses. These events posed direct threats to local social stability and public safety. During such circumstances, it is essential to formulate a comprehensive emergency rescue strategy grounded on a precise evaluation of the magnitude and consequences of the catastrophe. Furthermore, this plan should involve the efficient distribution of rescue resources and the implementation of communication systems to ensure successful evacuation and medical assistance. Implementing such relief programs can significantly reduce disaster-related losses and maintain the safety and stability of affected communities. Based on this, five different emergency rescue plans are proposed. These plans involve flood relief measures, emergency relocation of residents, distribution of relief materials, strategic deployment of rescue teams, and other key aspects, covering the following four main attributes:

- 1) Casualty level c_1 : This attribute measures the degree of loss of life in an accident or disaster.
- 2) Degree of economic loss c_2 : This attribute represents the immediate influence of an event on the economy, which can encompass consequences such as property destruction or a halt in output.
- 3) Traffic paralysis c_3 : This attribute quantifies the magnitude of the influence that disasters have on land and transportation infrastructure. The measurement quantifies the degree of disturbance to transport infrastructure, caused by natural catastrophes like floods and mudslides.
- 4) Precision facility resilience c_4 : This attribute is used to assess the resilience of critical infrastructure and vital services after a disaster. These advanced facilities may include medical centers, emergency services, crucial communications base stations, and other amenities.

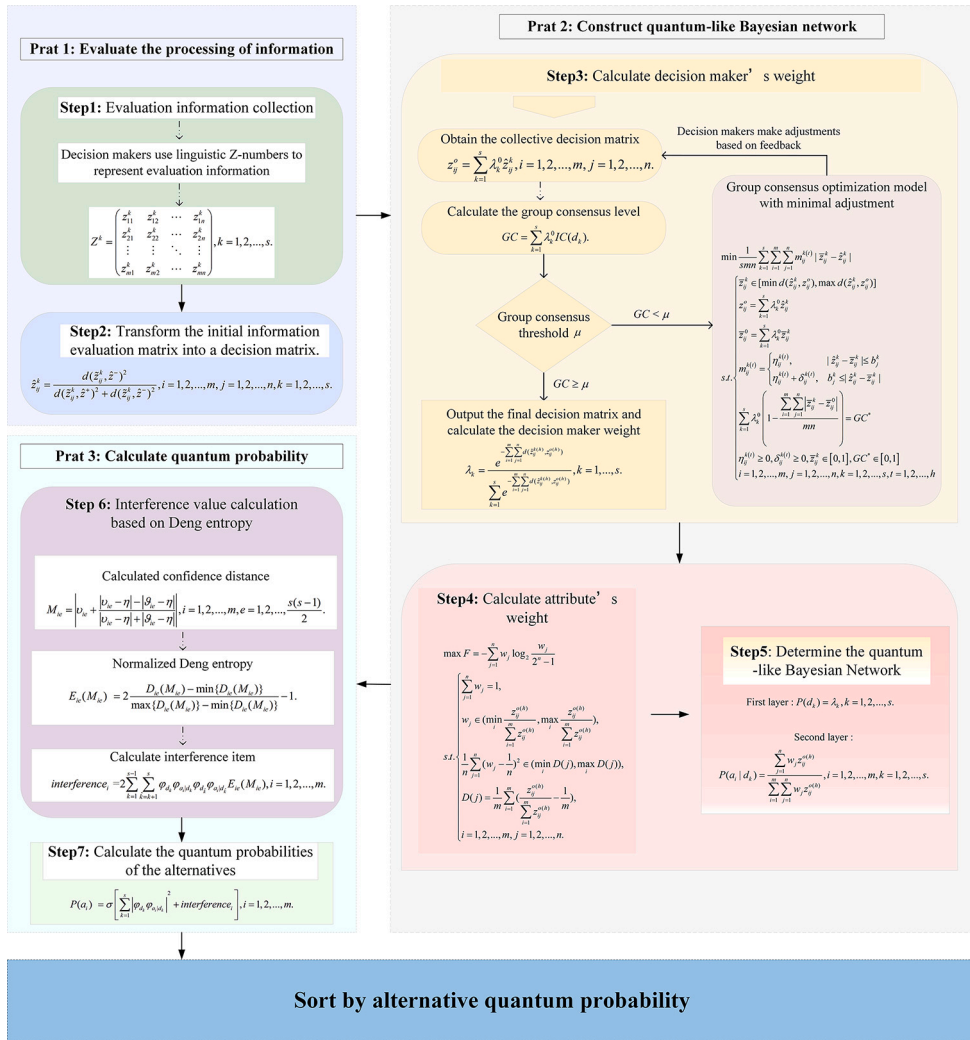


Fig. 3. Z-MAQGCDM model decision steps.

4.2. Decision steps

Step 1: Collect the evaluation information. The two linguistic sets of this case are:

- $A = \{A_0 : \text{very poor}, A_1 : \text{poor}, A_2 : \text{slightly poor}, A_3 : \text{fair},$
 $A_4 : \text{slightly good}, A_5 : \text{good}, A_6 : \text{very good}\}$
- $B = \{B_0 : \text{more unlikely}, B_1 : \text{unlikely}, B_2 : \text{fair},$
 $B_3 : \text{likely}, B_4 : \text{more likely}\}$

Thus, Table 2 displays the decision matrix for four decision-makers using the LZN of assessment information.

Step 2: Use Eq. (12) to convert $Z^k = (z_{ij}^k)_{m \times n}$ ($k = 1, 2, \dots, 4$) into a clear linguistic Z-numbers decision matrix $\hat{Z}^k$ ($k = 1, 2, \dots, 4$), where, $\rho_1 = \rho_2 = 0.5$, and result as shown in Table 3.

Step 3: Weight determination of decision-makers based on group consensus. Assuming that the initial weight set of each decision maker is $\Lambda = (0.25, 0.25, 0.25, 0.25)$ given by the top decision maker, the collective decision matrix $Z^o = (z_{ij}^o)_{m \times n}$ can be calculated by aggregating the decision matrix, as Table 4 illustrates.

In this instance, the threshold for group agreement is chosen at $\mu = 0.85$, and the bounded confidence matrix for each decision maker is shown in Table 5.

1) Round 1: Using Eqs. (13)-(15), the initial group consensus level $GC^{(0)} = 0.8062$ is obtained. Since $GC^{(0)} < 0.85$ it is suggested that policymakers adjust the decision matrix and solve Model (18) to get the first round of feedback decisions $\bar{Z}^{k(1)}$ ($k = 1, 2, \dots, 4$), as shown in Table 6.

Table 2Initial information evaluation matrix $Z^k (k = 1, 2, \dots, 4)$.

	c_1	c_2	c_3	c_4	c_1	c_2	c_3	c_4
	d_1				d_2			
a_1	(A_3, B_2)	(A_1, B_4)	(A_0, B_0)	(A_0, B_1)	(A_4, B_2)	(A_5, B_3)	(A_2, B_1)	(A_2, B_2)
a_2	(A_4, B_2)	(A_3, B_2)	(A_3, B_4)	(A_2, B_0)	(A_0, B_0)	(A_6, B_3)	(A_1, B_3)	(A_3, B_3)
a_3	(A_6, B_3)	(A_5, B_2)	(A_4, B_3)	(A_6, B_4)	(A_6, B_4)	(A_4, B_3)	(A_3, B_4)	(A_3, B_3)
a_4	(A_3, B_4)	(A_4, B_3)	(A_2, B_4)	(A_2, B_3)	(A_2, B_1)	(A_3, B_1)	(A_5, B_3)	(A_6, B_4)
	d_1				d_2			
a_1	(A_2, B_3)	(A_2, B_2)	(A_4, B_2)	(A_4, B_2)	(A_4, B_0)	(A_1, B_3)	(A_1, B_3)	(A_4, B_4)
a_2	(A_2, B_4)	(A_6, B_3)	(A_3, B_2)	(A_3, B_2)	(A_5, B_1)	(A_0, B_0)	(A_4, B_2)	(A_3, B_3)
a_3	(A_3, B_5)	(A_3, B_2)	(A_0, B_0)	(A_0, B_0)	(A_5, B_3)	(A_2, B_1)	(A_2, B_1)	(A_6, B_4)
a_4	(A_4, B_1)	(A_3, B_3)	(A_6, B_2)	(A_6, B_2)	(A_3, B_3)	(A_4, B_2)	(A_5, B_4)	(A_2, B_1)

Table 3Initial decision matrix $\hat{Z}^k (k = 1, 2, \dots, 4)$.

	c_1	c_2	c_3	c_4	c_1	c_2	c_3	c_4
	d_1				d_2			
a_1	0.6907	0.5968	0.0000	0.1667	0.8275	0.9693	0.3421	0.5354
a_2	0.8575	0.6907	0.8333	0.0714	0.0000	0.9898	0.5219	0.7887
a_3	0.9898	0.9132	0.9025	1.0000	1.0000	0.9025	0.8333	0.7887
a_4	0.8333	0.9025	0.7143	0.6506	0.3421	0.5000	0.9693	1.0000
	d_1				d_2			
a_1	0.6506	0.5354	0.8275	0.5000	0.2857	0.5219	0.5219	0.9286
a_2	0.7143	0.9025	0.6907	0.8275	0.7727	0.0000	0.8275	0.7887
a_3	0.8504	0.6907	0.0000	0.7887	0.9693	0.3421	0.3421	1.0000
a_4	0.6579	0.7887	0.9459	1.0000	0.7887	0.8275	0.9839	0.3421

Table 4Collective decision matrix $Z^{(0)}$.

	c_1	c_2	c_3	c_4
a_1	0.6136	0.6559	0.4229	0.5327
a_2	0.5786	0.6458	0.7184	0.6191
a_3	0.9524	0.7121	0.5195	0.8943
a_4	0.6555	0.7547	0.9033	0.7482

Table 5Bounded confidence for decision-makers
 $B^k (k = 1, 2, \dots, 4)$.

	c_1	c_2	c_3	c_4
d_1	0.15	0.10	0.20	0.30
d_2	0.05	0.15	0.10	0.16
d_3	0.30	0.45	0.21	0.04
d_4	0.20	0.15	0.20	0.25

Table 6First round feedback decision matrix $\tilde{Z}^{k(1)} (k = 1, 2, \dots, 4)$.

	c_1	c_2	c_3	c_4	c_1	c_2	c_3	c_4
	d_1				d_2			
a_1	0.6791	0.6027	0.0846	0.2765	0.8168	0.9223	0.3502	0.5350
a_2	0.7902	0.6862	0.8103	0.2357	0.0289	0.9382	0.5415	0.7616
a_3	0.9842	0.8931	0.8259	0.9683	0.9976	0.8739	0.8019	0.8056
a_4	0.8066	0.8877	0.7521	0.6799	0.3578	0.5382	0.9627	0.9597
	d_3				d_4			
a_1	0.6395	0.5896	0.7425	0.5013	0.3513	0.5420	0.5021	0.8296
a_2	0.6736	0.7870	0.6965	0.8192	0.7339	0.0969	0.8057	0.7463
a_3	0.8810	0.7003	0.1091	0.7929	0.9659	0.3976	0.3776	0.9736
a_4	0.6572	0.7734	0.9370	0.9899	0.7621	0.8166	0.9678	0.4436

Table 7
Second round adjustment decision matrix $\hat{Z}^{k(1)}(k = 1, 2, \dots, 4)$.

	c_1	c_2	c_3	c_4	c_1	c_2	c_3	c_4
	d_1				d_2			
a_1	0.6849	0.5968	0.2049	0.2326	0.7351	0.9016	0.3421	0.5354
a_2	0.8153	0.6907	0.8333	0.0714	0.2018	0.8963	0.5219	0.7887
a_3	0.9898	0.9132	0.9025	1.0000	0.9613	0.9025	0.8333	0.7887
a_4	0.8333	0.9025	0.7143	0.6506	0.5347	0.5000	0.9693	0.8916
	d_3				d_4			
a_1	0.6506	0.5354	0.8275	0.5000	0.3861	0.5219	0.5219	0.9286
a_2	0.7143	0.8849	0.6907	0.7249	0.7727	0.0463	0.7981	0.7887
a_3	0.8504	0.6907	0.2016	0.7887	0.9693	0.3421	0.3421	0.9462
a_4	0.6579	0.7887	0.9459	0.8805	0.7887	0.8275	0.9839	0.3719

Table 8
Second round feedback decision matrix $\bar{Z}^{k(2)}(k = 1, 2, \dots, 4)$.

	c_1	c_2	c_3	c_4	c_1	c_2	c_3	c_4
	d_1				d_2			
a_1	0.6743	0.6010	0.2587	0.3276	0.7291	0.8622	0.3553	0.5376
a_2	0.7869	0.6846	0.8088	0.2280	0.2230	0.8563	0.5408	0.7575
a_3	0.9827	0.8931	0.8360	0.9643	0.9604	0.8739	0.8070	0.8035
a_4	0.8139	0.8877	0.7521	0.6650	0.5431	0.5382	0.9627	0.8607
	d_3				d_4			
a_1	0.6397	0.5820	0.7533	0.5020	0.4317	0.5395	0.5123	0.8337
a_2	0.6878	0.7700	0.6950	0.7196	0.7434	0.1338	0.7807	0.7399
a_3	0.8781	0.7003	0.2789	0.7924	0.9640	0.3976	0.3877	0.9299
a_4	0.6716	0.7734	0.9370	0.8732	0.7717	0.8166	0.9678	0.4536

Table 9
Adjustment decision matrix $\hat{Z}^{k(2)}(k = 1, 2, \dots, 4)$.

	c_1	c_2	c_3	c_4	c_1	c_2	c_3	c_4
	d_1				d_2			
a_1	0.6849	0.5968	0.2049	0.2326	0.7351	0.8543	0.3421	0.5354
a_2	0.7404	0.6907	0.8333	0.0714	0.2018	0.8963	0.6179	0.7887
a_3	0.9898	0.9132	0.9025	1.0000	0.9613	0.9025	0.7963	0.7887
a_4	0.8333	0.9025	0.7516	0.6506	0.5347	0.5000	0.9514	0.8916
	d_3				d_4			
a_1	0.6506	0.5354	0.8275	0.5000	0.3861	0.5219	0.5219	0.7691
a_2	0.7143	0.8849	0.6907	0.7249	0.7727	0.0841	0.7981	0.7887
a_3	0.8804	0.6907	0.3149	0.7887	0.9693	0.3421	0.3421	0.9462
a_4	0.6579	0.7887	0.9459	0.8805	0.7887	0.8275	0.9839	0.7149

2) Round 2: According to the feedback suggestions, each decision maker adjusts the decision matrix to the decision matrix of Table 7.

Due to the current level of group consensus $GC^{(1)} = 0.8341$, which is still below the threshold, the data in Model (18) is updated again. This results in a new round of feedback matrices $\bar{Z}^{k(2)}(k = 1, 2, \dots, 4)$, as shown in Table 8.

3) Round 3: In this round, the adjusted decision matrix $\hat{Z}^{k(2)}(k = 1, 2, \dots, 4)$ is obtained based on the feedback suggestions, as shown in Table 9.

At this point, it is concluded that $GC^{(2)} = 0.8506 > \mu = 0.85$. This round of adjustment results in the group consensus level exceeding the predetermined threshold value; therefore, the adjustment process ends, and $\hat{Z}^{k(2)}(k = 1, 2, \dots, 4)$ is the final decision matrix for each decision maker.

Eq. (23) was applied to calculate the weight of decision-makers adjusted based on group consensus.

$$\Lambda = \{0.1682, 0.1929, 0.4586, 0.1803\}.$$

Step 4: The collective decision matrix is computed based on the revised decision matrix and decision-maker weights, as shown in Table 10.

Model (20) is solved to calculate the optimal attributes' weights.

$$W = \{0.2655, 0.2400, 0.2574, 0.2371\}.$$

Table 10
Collective decision matrix $Z^{(3)}$.

	c_1	c_2	c_3	c_4
a_1	0.6250	0.6048	0.5740	0.5104
a_2	0.6304	0.7100	0.7200	0.6388
a_3	0.9304	0.7061	0.5115	0.8526
a_4	0.6872	0.7591	0.9211	0.8141

Step 5: Determine the OLBN. The weights of the decision-makers are determined as the node probability in the first layer of the BN.

$$P(d_1) = 0.1682, \quad P(d_2) = 0.1929, \quad P(d_3) = 0.4586, \quad P(d_4) = 0.1803.$$

The conditional probability in the BN is obtained from Eq. (22).

$$\begin{aligned} P(a_1|d_1) &= 0.1567, & P(a_2|d_1) &= 0.2149, & P(a_3|d_1) &= 0.3442, & P(a_4|d_1) &= 0.2842, \\ P(a_1|d_2) &= 0.2188, & P(a_2|d_2) &= 0.2186, & P(a_3|d_2) &= 0.3072, & P(a_4|d_2) &= 0.2554, \\ P(a_1|d_3) &= 0.2206, & P(a_2|d_3) &= 0.2621, & P(a_3|d_3) &= 0.2327, & P(a_4|d_3) &= 0.2846, \\ P(a_1|d_4) &= 0.2059, & P(a_2|d_4) &= 0.2336, & P(a_3|d_4) &= 0.2464, & P(a_4|d_4) &= 0.3141. \end{aligned}$$

The probability of each alternative is obtained:

$$P(a_1) = 0.2069, \quad P(a_2) = 0.2406, \quad P(a_3) = 0.2683, \quad P(a_4) = 0.2842.$$

The alternative is ranked according to the probability value: $a_4 > a_3 > a_2 > a_1$.

Step 6: Interference value calculation based on Deng entropy. The standardized Deng entropy is calculated as follows:

$$E(M) = \begin{bmatrix} -0.1018 & -0.7503 & -0.3474 & -0.0029 & -0.4710 & -0.0985 \\ -0.4403 & 0.1079 & -0.4436 & 0.3163 & -0.8091 & 0.4489 \\ -0.2767 & 0.8623 & 0.5377 & 0.9347 & 0.6966 & 0.3267 \\ -0.6238 & 0.6017 & 0.2378 & 0.6494 & 0.1592 & 1.0000 \end{bmatrix}.$$

Step 7: Consider alternative probability calculations for quantum interference effects. The quantum probability of each alternative is obtained using Eq. (31).

$$P(a_1) = 0.0318, \quad P(a_2) = 0.1411, \quad P(a_3) = 0.4306, \quad P(a_4) = 0.3964.$$

The alternative is ranked according to the quantum probability value: $a_3 > a_4 > a_2 > a_1$.

4.3. Sensitivity analysis

1) The influence of group consensus level on the weights of decision makers

The establishment of the group consensus threshold is vital for the functioning and precision of the model. Different thresholds will directly affect the decision matrix of each decision maker, resulting in changes in their choices and judgments, which in turn affect the overall model output. In this model, altering the group consensus threshold will not only change the decision matrix of each decision maker but also further impact the weight allocation of decision makers, as shown in Fig. 4.

As shown in Fig. 4, a higher group consensus threshold may require more decision-makers to reach an agreement, thereby improving the accuracy and consistency of the overall decision. However, this can also result in slower decision-making. Conversely, a lower group consensus threshold can enhance decision efficiency but may reduce consistency and accuracy.

2) Verification of the effectiveness and robustness of the proposed model

The validity and robustness of the model are verified in this paper. In order to verify the validity of the model, this paper randomly generates an initial decision matrix of different dimensions into the model, assumes that decision makers accept the model output feedback suggestion matrix, and iterates repeatedly to observe the changes in the group consensus level. In addition, the influence of different bounded confidence levels on the consensus level of decision makers is also considered. According to literature [15], the number of iterations is set to 5 in this paper, in which the random decision matrix is generated by the “rand” function of matlab, and the specific results are shown in Fig. 5.

As can be seen from Fig. 5, decision maker confidence has a significant impact on the group consensus level. With the increase of the number of iteration rounds, the consensus degree of each confidence level shows an increasing trend, which indicates that the model has a strong dynamic adjustment ability under different confidence conditions. However, groups of decision makers with higher confidence (b_j^k ($j = 1, 2, \dots, 5, k = 1, 2, \dots, 4$) ≥ 0.7) were able to reach a high level of consensus faster and stabilize in fewer rounds, which correlated with their high trust in feedback. In contrast, decision makers with lower confidence (b_j^k ($j = 1, 2, \dots, 5, k = 1, 2, \dots, 4$) ≤ 0.3)

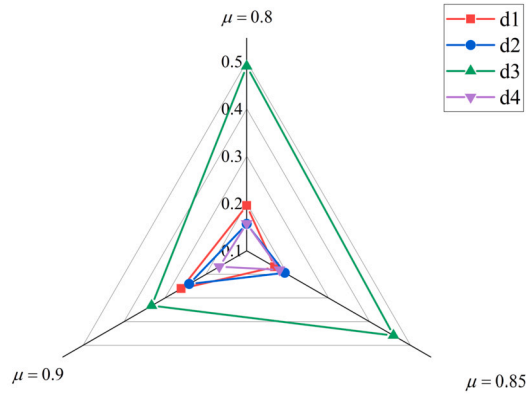


Fig. 4. Weight changes of decision makers under different group consensus thresholds.

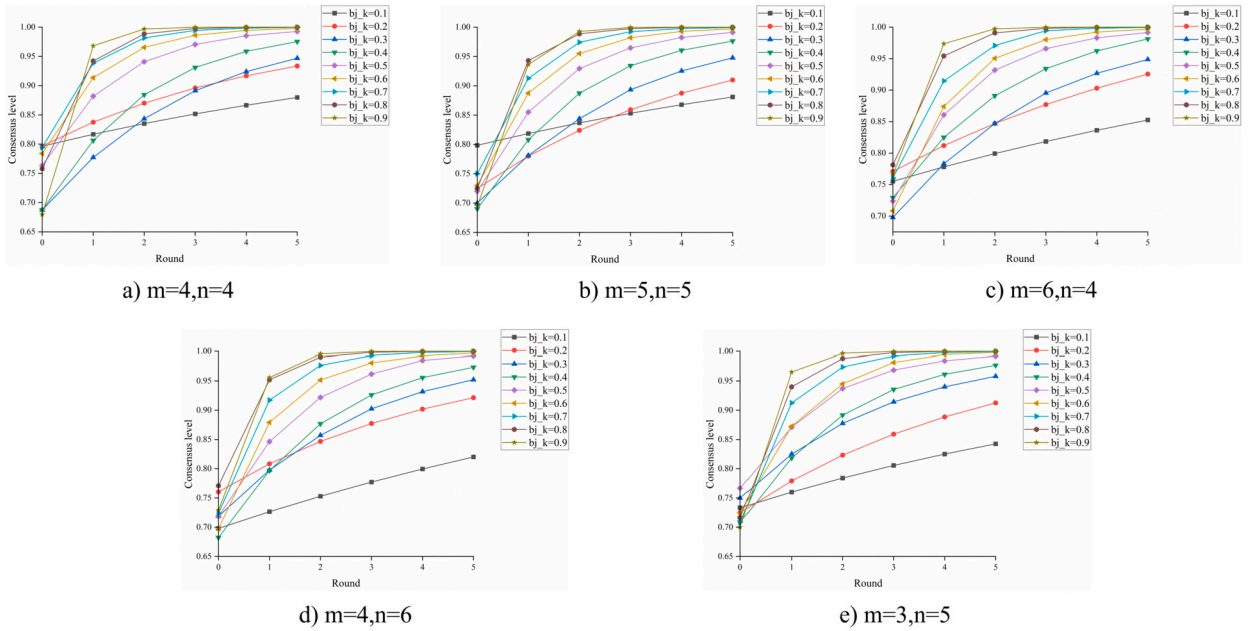


Fig. 5. Group consensus level variation of decision matrix with different dimensions.

require more iterations to gradually improve the consensus level, and their final consensus level is lower than that of the high confidence group, indicating that they are slower to respond to feedback.

From the matrix comparison of different dimensions, although the initial consensus level is different, the influence mode of confidence on consensus reaching remains consistent. No matter in small-scale decision making scenarios ($m = 4, n = 4$) or large-scale scenarios ($m = 6, n = 4$), decision makers with higher confidence can always rapidly improve the consensus level, indicating that the model has strong robustness in different scenarios. This result reflects that in the multi-attribute group decision-making process, confidence, as an important factor for decision makers to adjust their evaluation and accept feedback, directly affects the process of consensus formation.

In summary, the confidence degree of decision makers has a significant impact on the results of the group consensus model. The higher the confidence, the easier it is for group decision makers to reach a high level of consensus in a shorter iteration round, and the final consensus level is higher. When the confidence is low, the group needs more iteration rounds to gradually improve the consensus level, and the final consensus level is relatively low. This shows that the model has good robustness under different confidence levels and can reflect the different response degrees of decision makers in feedback adjustment. At the same time, the model has the same performance to the change of confidence in different scale decision problems, which verifies its adaptability to different input data and decision environment.

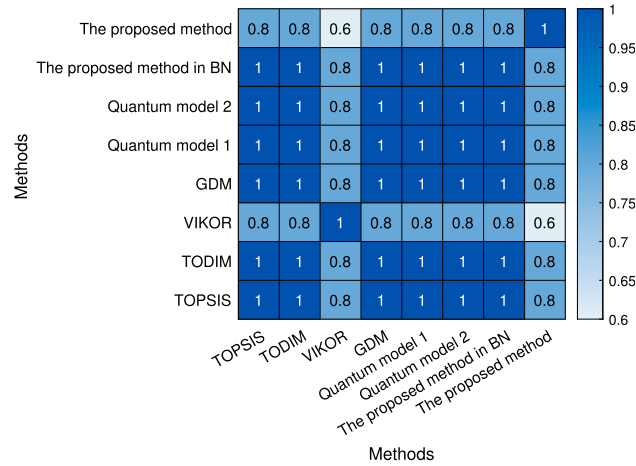


Fig. 6. Spearman rank correlation of different methods.

Table 11
Comparison of ranking results of different models.

Methods	Ranking indices	Final ranking orders
TOPSIS [47]	$C_1^* = 0.1086, C_2^* = 0.4040,$ $C_3^* = 0.4780, C_4^* = 0.7286.$	$a_4 > a_3 > a_2 > a_1$
TODIM [48]	$\phi(a_1) = -3.9874, \phi(a_2) = -1.3462,$ $\phi(a_3) = 0.0963, \phi(a_4) = 0.9438.$	$a_4 > a_3 > a_2 > a_1$
VIKOR [49]	$Q_1 = 1.0000, Q_2 = 0.7151,$ $Q_3 = 0.4956, Q_4 = 0.0000.$	$a_4 > a_3 > a_1 > a_2$
GDM [15]	$Q_1 = 0.5804, Q_2 = 0.6752,$ $Q_3 = 0.7510, Q_4 = 0.7955.$	$a_4 > a_3 > a_2 > a_1$
Quantum model 1 [50]	$P(a_1) = 0.2000, P(a_2) = 0.2402,$ $P(a_3) = 0.2747, P(a_4) = 0.2885.$	$a_4 > a_3 > a_2 > a_1$
Quantum model 2 [34]	$P(a_1) = 0.2093, P(a_2) = 0.2438,$ $P(a_3) = 0.2670, P(a_4) = 0.2799.$	$a_4 > a_3 > a_2 > a_1$
The proposed method in BN	$P(a_1) = 0.2069, P(a_2) = 0.2406,$ $P(a_3) = 0.2683, P(a_4) = 0.2842.$	$a_4 > a_3 > a_2 > a_1$
The proposed method	$P(a_1) = 0.0318, P(a_2) = 0.1411,$ $P(a_3) = 0.4306, P(a_4) = 0.3964.$	$a_3 > a_4 > a_2 > a_1$

4.4. Comparative analysis

In order to evaluate the effectiveness of the proposed method, different decision models are compared and analyzed. These models are compared directly or indirectly to verify the effect of the model in this paper. Fig. 6 shows the Spearman ranking correlation between the paper's model and other methods.

As can be seen from Fig. 6, when the model is converted into classical probability, the ranking correlation coefficient increases significantly, indicating a strong correlation between ranking results. This further verifies the effectiveness and adaptability of the proposed method in the classical decision environment.

The conclusive ranking outcomes achieved by various approaches are shown in Table 11. The rescue alternative rankings derived by the suggested method in Table 11 deviate considerably from those of the traditional decision model. For example, the optimal alternatives obtained by the TOPSIS [47] method, TODIM [48] method, VIKOR [49] method, and GDM [15] model are all a_4 . When the proposed model is utilized in the BN, the optimal plan is also a_4 . This indicates that the ranking outcomes of the proposed model are consistent with those of conventional models when the influence of decision-makers is not taken into account. Consequently, this confirms the soundness of the proposed model. However, when considering the quantum interference among decision-makers, the results of this model indicate that a_3 is the best alternative. The discrepancy arises from the omission of the interference effect among decision makers in the standard paradigm, resulting in subjective influence among decision makers when assessing alternative options.

In contrast, compared to the quantum model 1 [50], the optimal alternative is a_4 . This discrepancy arises because the quantum model cannot measure the specific angle of the phase angle, affecting the accuracy of the result. However, the method proposed in this model considers the concrete value of the interference rather than limiting all phase angles to a fixed angle. Consequently, the results of this model are more consistent with real-world complexities. The ranking results of quantum model 2 [34] also indicate

that alternative a_4 is the optimal choice. This outcome primarily arises from the model's inability to objectively measure attributes' weights during analysis and its oversight of decision-makers' need to adjust confidence intervals. Consequently, this leads to a partial misrepresentation of the actual situation. To address this, this paper introduces Deng entropy to determine attributes' weights and incorporates group consensus adjustments based on compensation costs and confidence intervals. By effectively capturing the fuzziness in information, this approach manages uncertainty and fuzziness more efficiently, resulting in ranking outcomes that better align with real-world needs.

5. Conclusion

In this paper, a multi-attribute fuzzy quantum group consensus model is proposed to solve the problem of information uncertainty and opinion interference among decision makers during emergency rescue of meteorological disasters. By introducing LZNs to represent fuzzy information in complex decision making, the model constructs a group consensus model based on the principle of minimum adjustment cost. The decision-makers adjust their opinions according to the feedback suggestion matrix of the model output, and consider the influence of confidence level on the efficiency of group consensus. The weight of the decision maker is determined by the distance between the final decision matrix and the collective decision matrix. In addition, a programming model is constructed based on the maximum entropy principle to solve the optimal attribute weights. At the same time, a quantum-like Bayesian network is introduced to deal with the interference effect between decision makers and calculate the quantum probability of each alternative scheme, where the interference angle is measured by Deng entropy. The results show that the model can significantly improve the consensus level of group decision making under different confidence backgrounds and multidimensional matrix inputs, and can effectively deal with interference effects among decision makers, and has strong robustness and wide applicability.

However, this paper still has the following limitations: 1) The model fails to adequately deal with the effect of large-scale group interference. With the increase of the number of decision makers and attribute dimension, the computational complexity of the model increases significantly. 2) The influence of decision makers' mental states on group consensus efficiency under complex uncertain situations is not fully considered. These problems need to be further discussed and improved in the follow-up study.

Based on the research results of this paper, future research directions can focus on the following solid aspects: 1) With the increase of the number of decision-makers and attribute dimensions, the computational complexity of the existing model increases significantly. Future studies may explore ways to reduce the computational complexity of large decision groups by optimizing algorithms or introducing parallel computing techniques. 2) The psychological factors of decision makers can be incorporated into the model in combination with the behavioral decision-making theory, so as to further improve the prediction accuracy of the model for decision-making behavior and enhance its adaptability in complex decision-making situations. 3) Further explore how to introduce dynamic interference effect into the model, so that the interference relationship between decision makers can be dynamically adjusted with the advance of time or decision-making process, and enhance the flexibility and accuracy of the model.

CRediT authorship contribution statement

Yizhao Xu: Writing – original draft, Investigation, Data curation. **Shuli Yan:** Writing – review & editing, Methodology, Conceptualization. **Ignacio Javier Pérez:** Validation, Supervision, Conceptualization. **Francisco Javier Cabrerizo:** Supervision, Conceptualization.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, and we do not have any possible conflicts of interest.

Acknowledgement

This work is funded by the Social Science Fund Project of Jiangsu Province (23GLB006), the National Natural Science Foundation of China (72371137, 71801085), Jiangsu University Philosophy and Social Science project (2022SJYB0187), the National Scholarship Foundation of China (202209045002), the grant PID2022-139297OB-I00 funded by MICIU/AEI/10.13039/501100011033 and by ERDF/EU, and by the project C-ING-165-UGR23, co-funded by the Regional Ministry of University, Research and Innovation and by the European Union under the Andalusia ERDF Programme 2021-2027.

Data availability

The authors do not have permission to share data.

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